CMAN:=

Given data

Peak compute = 10TFLOPS (peak point)

P_{peak} = 10,000 GFLOPS

∴ 1 TFlops = 1000 GFlops

Dram Bandwidth = 320 GB/s → B = 320 GB/s

Ridge Point = $\frac{P_{peak}}{B} = \frac{10{,}000}{320}$ = 31.25 Flop/byte

[∴ 31.25 Flop/byte (Below) → Memory bound DRAM
31.25 Flop/byte (Above) → compute bound (peak FLOPS)]

Tasks:=

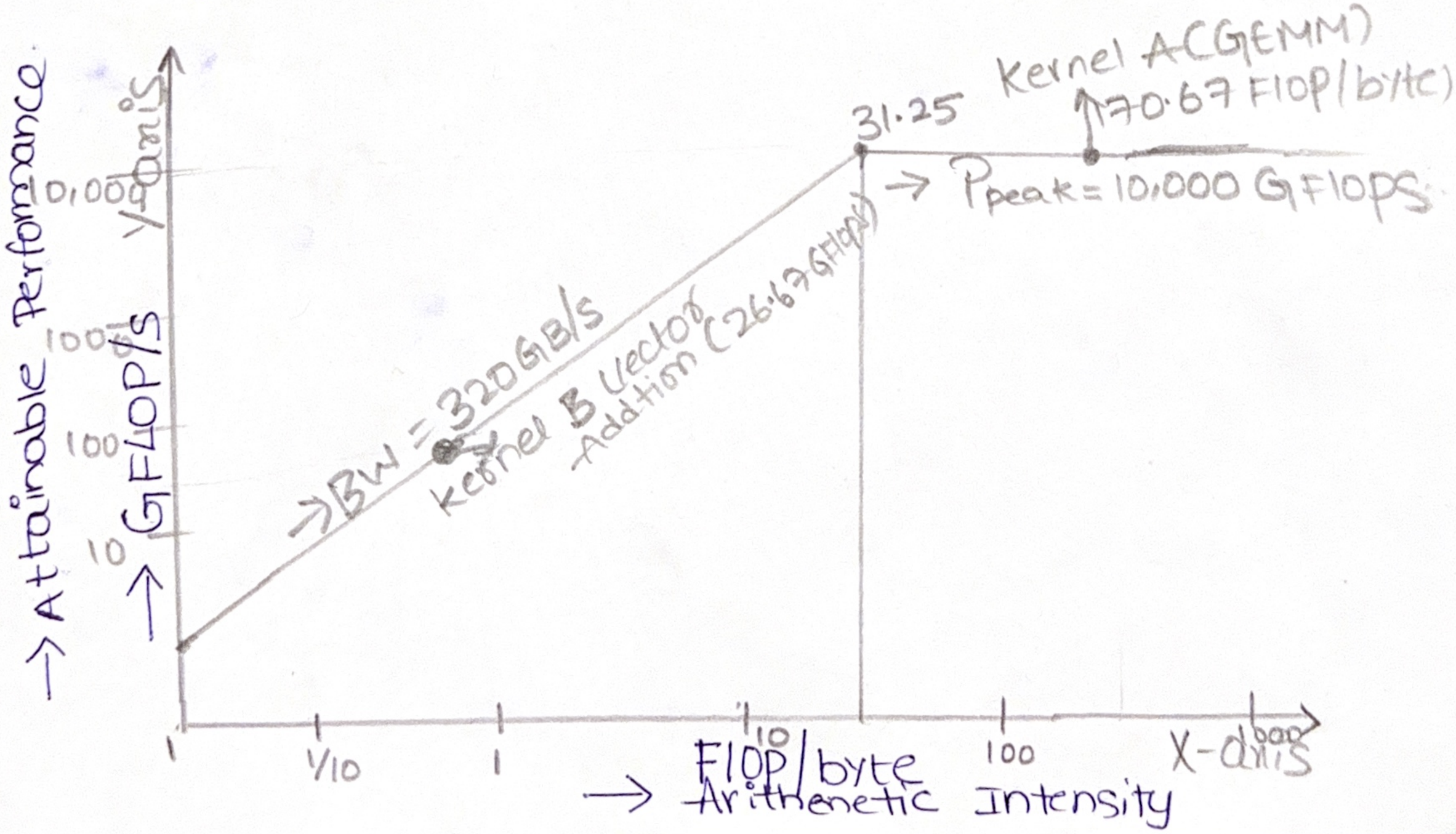
① Bandwidth limited

P = Arithmetic Intensity × B

compute limited

P = P_{peak} = 10,000 GFlops

Ridge Point = (31.25 Flop/byte , 10,000 GFlops)



② Kernel A: Dense GEMM

Flops $= 2 \times N^3$

$= 2 \times (1024)^3$

Flops $= 2{,}147{,}483{,}648$ Flops.

Bytes = One Matrix $= 1024 \times 1024 \times 4 = 4{,}194{,}304$ bytes

3 Matrices $= 3 \times 4{,}194{,}304 = 12{,}582{,}912$ bytes

Arithmetic Intensity (AI)

$$AI = \frac{\text{Flops}}{\text{Bytes}} = \frac{2{,}147{,}483{,}648}{12{,}582{,}912} = 170.67 \text{ Flop/byte}$$

$$\boxed{AI = 170.67 \text{ Flop/byte}}$$

$170.67 > 31.25$ (Compute bond, It's above)

Attainable performance $= 10{,}000$ GFlops/s.

③ Kernel B: Vector addition

Flops = One Matrix $= 1024 \times 1024 \times 4 = 4{,}194{,}304$ bytes

Bytes = for One vector $= 4{,}194{,}304 \times 4 = 16{,}777{,}216$ bytes

3 vector's $= 3 \times 16{,}777{,}216 = 50{,}331{,}648$ bytes

Arithmetic Intensity (AI) $:= \dfrac{\text{Flops}}{\text{Bytes}}$

$$AI = \frac{4{,}194{,}304}{50{,}331{,}648} = 0.083 \text{ Flop/byte}$$

$$\boxed{AI = 0.083 \text{ Flop/byte}}$$

$0.083 < 31.25$ (It's below the ridge point, Memory bound)

Attainable performance $= 320 \times 0.083 = 26.67$ GFlop/s

④ Kernel A : GEMM

a) Bound : compute Bound (170.67 > 31.25)
b) Attainable : 10,000 GFlop/s (hits compute ceiling)
c) Recommendation: We can add tensor cores and more MAC units.

GEMM is compute bound, so more FP32 throughput directly raises performance. More bandwidth would not help here.

Kernel B : Vector addition

a) Bound : Memory Bound (0.083 < 31.25)
b) Attainable : 320 x 0.083 = 26.67 GFlop/s
c) Recommendation : switch to High Bandwidth memory.

Vector add does 1 FLOP per 12 bytes, Compute Units sit idle waiting for data. More ALUs would not help, bandwidth is the bottleneck.